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12

Half-Yearly Examination - 2025
MATHEMATICS

Reg. No.

Time : 3.00 hrs. Max. Marks : 90

12-A, 02

PART – I

20 x 1 = 20

(i) All questions are compulsory.
(ii) Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer.

1. If $A = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ 2 & 0 \end{bmatrix}$ then $|adj\ (AB)| =$ a) –40 b) –80 c) –60 d) –20

2. If $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ and $A(adj\ A) = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$, then k = a) 0 b) sin c) cos d) 1

3. If $|z_1| = 1$, $|z_2| = 2$, $|z_3| = 3$ and $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 12$, then the value of $|z_1 + z_2 + z_3|$ is a) 1 b) 2 c) 3 d) 4

4. If α and β are the roots of $x^2 + x + 1 = 0$, then $\alpha^{2020} + \beta^{2020}$ is a) –2 b) –1 c) 1 d) 2

5. The polynomial $x^3 - kx^2 + 9x$ has three real zeros if and only if, k satisfies a) $|k| \le 6$ b) $k = 0$ c) $|k| > 6$ d) $|k| \ge 6$

6. If $\cot^{-1} x = \frac{2\pi}{5}$ for some $x \in R$, the value of $\tan^{-1} x$ is a) $-\frac{\pi}{10}$ b) $\frac{\pi}{5}$ c) $\frac{\pi}{10}$ d) $-\frac{\pi}{5}$

7. If $\sin^{-1} x + \cot^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{2}$, then x is equal to a) $\frac{1}{2}$ b) $\frac{1}{\sqrt{5}}$ c) $\frac{2}{\sqrt{5}}$ d) $\frac{\sqrt{3}}{2}$

8. The radius of the circle $3x^2 + by^2 + 4bx - 6by + b^2 = 0$ is a) 1 b) 3 c) $\sqrt{10}$ d) $\sqrt{11}$

9. The eccentricity of the ellipse $(x-3)^2 + (y-4)^2 = \frac{y^2}{9}$ is a) $\frac{\sqrt{3}}{2}$ b) $\frac{1}{3}$ c) $\frac{1}{3\sqrt{2}}$ d) $\frac{1}{\sqrt{3}}$

10. If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar, non-zero vectors such that $[\vec{a}, \vec{b}, \vec{c}] = 3$, then $\{[\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}]\}^2$ is equal to
a) 81 b) 9 c) 27 d) 18

11. The distance between the planes $x + 2y + 3z + 7 = 0$ and $2x + 4y + 6z + 7 = 0$ is a) $\frac{\sqrt{7}}{2\sqrt{2}}$ b) $\frac{7}{2}$ c) $\frac{\sqrt{7}}{2}$ d) $\frac{7}{2\sqrt{2}}$

12. Angle between $y^2 = x$ and $x^2 = y$ at the origin is a) $\tan^{-1}\frac{3}{4}$ b) $\tan^{-1}\left(\frac{4}{3}\right)$ c) $\frac{\pi}{2}$ d) $\frac{\pi}{4}$

13. The point of inflection of the curve $y = (x-1)^3$ is a) (0, 0) b) (0, 1) c) (1, 0) d) (1, 1)

14. If we measure the side of a cube to be 4 cm with an error of 0.1 cm, then the error in our calculation of the volume is
a) 0.4 cu.cm b) 0.45 cu.cm c) 2 cu.cm d) 4.8 cu.cm

15. The value of $\int_0^1 x(1-x)^{99}\,dx$ is a) $\frac{1}{11000}$ b) $\frac{1}{10100}$ c) $\frac{1}{10010}$ d) $\frac{1}{10001}$

16. If $\int_0^a \frac{1}{4+x^2}dx = \frac{\pi}{8}$ then a is a) 4 b) 1 c) 3 d) 2

17. The solution of $\frac{dy}{dx} + p(x)y = 0$ is a) $y = ce^{\int pdx}$ b) $y = ce^{-\int pdx}$ c) $x = ce^{-\int pdy}$ d) $x = ce^{\int pdy}$

18. A random variable X has binomial distribution with n = 25 and p = 0.8 then standard deviation of X is
a) 6 b) 4 c) 3 d) 2

19. The probability mass function of a random variable is defined as

x	–2	–1	0	1	2
f(x)	k	2k	3k	4k	5k

Then E(X) is equal to : a) $\frac{1}{15}$ b) $\frac{1}{10}$ c) $\frac{1}{3}$ d) $\frac{2}{3}$

20. Subtraction is not a binary operation in a) R b) Z c) N d) Q

PART - II

Answer any seven questions. Question No.30 is compulsory. 7 x 2 = 14

21. Prove that if A is symmetric, prove that adj A is also symmetric.

22. Write in polar form of the complex numbers $3 - i\sqrt{3}$

23. Find the value of $\cos\left(\cos^{-1}\left(\frac{4}{5}\right) + \sin^{-1}\left(\frac{4}{5}\right)\right)$

24. Find centre and radius of the circle $x^2 + y^2 + 6x - 4y + 4 = 0$

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25. A particle is acted upon by the forces $3\hat{i}-2\hat{j}+2\hat{k}$ and $2\hat{i}+\hat{j}-\hat{k}$ is displaced from the point ( ) to the point $(4,-1,\lambda)$. If the work done by the forces is 16 units, find the value of λ.

26. Evaluate $\lim_{x\to 0}\frac{1-\cos x}{x^2}$

27. A sphere is made of ice having radius 10 cm. Its radius decreases from 10 cm to 9.8 cm. Find approximations for change in the volume.

28. Evaluate $\int_3^4 \frac{dx}{x^2-4}$

29. Let $A=\begin{bmatrix}0 & 1\\ 1 & 1\end{bmatrix}$, $B=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$ be any two boolean matrices of the same type. Find $A\vee B$ and $A\wedge B$

30. Show that the differential equation of the family of the curve $y = Ae^x+Be^{-x}$ where A and B are arbitrary constants, is $\frac{d^2y}{dx^2}-y=0$.

PART - III

Answer any seven questions. Question No.40 is compulsory. **7 x 3 = 21**

31. Solve the following system of linear equations by matrix inversion method : $2x - y = 8$, $3x + 2y = -2$.

32. Show that (i) $(2+i\sqrt{3})^{10}+(2-i\sqrt{3})^{10}$ is real.

33. Find the sum of squares of roots of the equation $2x^4 - 8x^3 + 6x^2 - 3 = 0$

34. Find the equation of the ellipse with foci $(\pm 2, 0)$, vertices $(\pm 3, 0)$.

35. If $\hat{a},\hat{b},\hat{c}$ are three unit vectors such that $\hat{b}$ and $\hat{c}$ are non-parallel and $\hat{a}\times(\hat{b}\times\hat{c})=\frac{1}{2}\hat{b}$, find the angle between $\hat{a}$ and $\hat{c}$.

36. Expand $\log(1 + x)$ as a Maclaurin's series upto 4 non-zero terms for $-1 < x \le 1$.

37. Evaluate $\int_0^1 x^3(1-x)^4\,dx$

38. Compute $P(X = k)$ for the binomial distribution, $B(n, p)$ where $n = 7$, $p = \frac{1}{3}$, $k = -3$.

39. Prove that $q \to p \equiv \neg p \to \neg q$

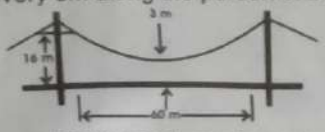
40. Show that $[\vec{a}\times\vec{b},\ \vec{b}\times\vec{c},\ \vec{c}\times\vec{a}] = [\vec{a}\ \vec{b}\ \vec{c}]^2$

PART - IV

Answer all the questions. **7 x 5 = 35**

41. a) A boy is walking along the path $y = ax^2 + bx + c$ through the points $(-6, 8)$, $(-2, -12)$, and $(3, 8)$. He wants to meet his friend at P(7, 60). Will he meet his friend? (Use Gaussian elimination method). **(OR)**

41. b) Parabolic cable of a 60m portion of the roadbed of a suspension bridge are positioned as shown below. Vertical Cables are to be spaced every 6m along the portion of the roadbed. Calculate the lengths of first two of these vertical cables from the vertex.



42. a) A camera is accidently knocked of an edge of cliff 400 ft. high. The camera falls a distance of $s = -16t^2$ in t seconds. Show that the camera hits the ground when t = 5 seconds and also prove that the velocity when it hits the ground is –160 ft/sec. **(OR)**

42. b) Solve the equation $6x^4 - 5x^3 - 38x^2 - 5x + 6 = 0$ if it is known that $\frac{1}{3}$ is a solution.

43. a) Solve the equation $z^3 + 8i = 0$, where $z \in C$. **(OR)**

43. b) Find the area of the region bounded by the line $7x - 5y = 35$, x - axis and the lines $x = -2$ and $x = 3$.

44. a) Find the parametric vector, non-parametric vector and Cartesian form of the equation of the plane passing through the three non-collinear points (3, 6, –2), (–1, –2, 6), and (6, 4, –2). **(OR)**

44. b) If $z = x + iy$ and $\arg\left(\frac{z-i}{z+2}\right)=\frac{\pi}{4}$, show that $x^2 + y^2 + 3x - 3y + 2 = 0$.

45. a) If $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \pi$, show that $x + y + z = xyz$. **(OR)**

45. b) If $u(x, y) = \frac{x^2+y^2}{\sqrt{x+y}}$, prove that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\frac{3}{2}u$.

46. a) A tank contains 1000 litres of water in which 100 gram of salt is dissolved. Brine (Brine is a high-concentration solution of salt (usually sodium chloride) in water) runs in a rate of 10 litres per minute, and each litre contains 5 grams of dissolved salt. The mixture of the tank is kept uniform by stirring. Brine runs out at 10 litres per minute. Find the amount of salt at any time? **(OR)**

46. b) Identify the type of conic and find centre, foci, vertices and directrices of the equation. $18x^2 + 12y^2 - 144x + 48y + 120 = 0$

47. a) Prove by vertex method that $\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$. **(OR)**

47. b) A random variable X has the following probability mass function.

x	1	2	3	4	5
f(x)	k^2	$2k^2$	$3K^2$	$2K$	$3K$

Find (i) the value of k (ii) $P(2 \le X < 5)$ (iii) $P(3 < X)$

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